

02 Slide structure

Visual Presentation: Critical Assessment of "Rethinking Financial Distress in Public Healthcare"

SLIDE 1: TITLE SLIDE

Critical Assessment of Academic Research

"Rethinking Financial Distress in Public Healthcare: Evidence from Greek Hospitals' Governance Attributes"

Author: Stefanos Karakolias (2025)
Journal: *Frontiers in Public Health*
Evaluator: [Your Name]
Course: Advanced Corporate Finance
Date: [Presentation Date]

SLIDE 2: THE RESEARCH AT A GLANCE

What the Paper Does

- Applies **Altman's Z"-score model** to 90 Greek public hospitals (2022)
- Tests relationship between **governance attributes** and financial distress
- Examines: Location, Size, Specialty, CEO Gender

Central Finding

- All hospitals have Z-scores > 5.5 (mean: 15.7) → "Financially Viable"
- **BUT:** Paper argues this is misleading due to:
 - €1.9B annual government subsidies
 - 1,485-day receivables collection period
 - Rising overdue liabilities (€344M → €1,164M in 5 years)

Key Claim

"Financial distress requires redefinition in the context of public entities, since their closure is not a core strategy"

SLIDE 3: EVALUATION FRAMEWORK - EIGHT DIMENSIONS

Dimension	Weight	Rating	Grade
I. Theoretical Foundation	25%	★ ★ ★ ☆ ☆	Weak
II. Methodological Rigor	30%	★ ★ ★ ½ ☆ ☆	Weak-Adequate
III. Results & Interpretation	20%	★ ★ ★ ★ ☆ ☆	Adequate
IV. Contribution to Knowledge	15%	★ ★ ★ ½ ☆ ☆	Weak-Adequate
V. Course Alignment	5%	★ ★ ★ ★ ☆ ☆	Adequate
VI. Practical Implications	3%	★ ★ ★ ★ ½ ☆	Adequate-Good
VII. Limitations & Future Research	2%	★ ★ ★ ★ ☆ ☆	Adequate
VIII. Writing Quality	-	★ ★ ★ ★ ★ ☆	Good

OVERALL GRADE: C+ / B- (70-75/100)

Classification: Adequate empirical work with significant theoretical and methodological limitations

SLIDE 4: ASSESSMENT METHODOLOGY

Evaluation Approach

Core Questions for Each Dimension:

1. Does it meet **Advanced Corporate Finance** standards?

- 2. Does it align with **syllabus learning objectives**?
- 3. Could it be published in a **Q1 management/finance journal**?
- 4. Does it advance **theory, method, or practice**?

 Reference Points

- **Syllabus topics:** Capital Structure, Corporate Governance, M&A, International Finance, Real Options
- **Methodological standards:** Causal inference, multivariate analysis, identification strategies
- **Theoretical standards:** Model development, testable hypotheses, theoretical extension
- **Benchmark papers:** Gemini, ChatGPT, and Claude research proposals on VBHC

SLIDE 5: THEORETICAL FOUNDATION - Critical Assessment

 MAJOR WEAKNESS: Theoretical Underdevelopment

The Core Problem

Paper's Logic:

- 1. Apply bankruptcy prediction model (Z-score)
- 2. To population with ZERO bankruptcies (Greek public hospitals)
- 3. Find model shows no distress
- 4. Conclude: Model is wrong, need new theory
- 5. BUT: No new theory provided! ⚠️

What's Missing

Missing Element	Why It Matters	Impact
Formal definition of "financial distress" in public entities	Can't measure what you can't define	Construct invalidity
Alternative theoretical framework beyond critique of Z-scores	Promised but not delivered	No theoretical contribution
Engagement with soft budget constraint theory (Kornai)	Explains why traditional models fail	Missed theoretical foundation
Application of capital structure theory (pecking order, trade-off)	Could explain observed financing patterns	Missed explanatory opportunity

SLIDE 6: THEORETICAL FOUNDATION - Syllabus Disconnect

 Capital Structure Theory Coverage

Syllabus Requires:

- Tradeoff vs. pecking order theories
- Dynamic models of capital structure
- Managerial discretion and CS
- Testing CS theories

Paper Provides:

-  Describes extreme capital structure (equity/liability = 10.9:1)
-  No theoretical explanation using CS theory
-  No test of competing theories
-  No engagement with dynamic models

 Corporate Governance Theory Coverage

Syllabus Requires:

- Board of Directors: theories and evidence
- Executive compensation
- Governance and equity returns

Paper Provides:

-  Tests CEO gender effect on performance
-  Atheoretical governance variables (gender as binary, size as proxy)
-  No data on boards, compensation, or governance mechanisms
-  No agency-theoretic model of public hospital governance

SLIDE 7: THEORETICAL CONTRIBUTION - What's Actually Offered

💡 The One Genuine Insight: "Transferred Distress"

Concept: Public hospitals don't experience distress themselves (state bailout), but **transfer it to stakeholders**:

```
Traditional Financial Distress (Private)
Firm → Bankruptcy → Stakeholders lose
```

```
"Transferred Distress" (Public)
Entity survives ← State absorbs costs
↓
Suppliers: 1,485-day payment delays
Taxpayers: €1.9B annual subsidies
Patients: Potential quality deterioration
```

Problem: This insight remains **underdeveloped**

- No formal model
- No testable predictions
- No measurement framework

Grade: WEAK (2/5) - Identifies problem, doesn't solve it

SLIDE 8: METHODOLOGICAL RIGOR - Three Fatal Flaws

🚨 **FLAW 1: Wrong Tool for the Job**

Altman's Z"-Score designed to predict:

- **Bankruptcy probability** in next 1-2 years
- For **private service firms**
- In **competitive markets**
- Validated on sample with actual failures

Applied to:

- **Public hospitals** where bankruptcy is impossible by design
- **Zero observed failures** in sample
- **Soft budget constraints** with continuous state support

| "Although no bankruptcies or closures of these entities have been reported to date" (p. 2)

Result: No validation criterion - Can't test if model works because outcome never occurs

SLIDE 9: METHODOLOGICAL RIGOR - Flaws Continued

🚨 **FLAW 2: Cross-Sectional Design for Dynamic Phenomenon**

Financial Distress = Process of Deterioration Over Time

```
Proper Study Design:
Time 0: Measure Z-scores
Time 1-3: Observe which hospitals experience crisis
Analysis: Do low Z-scores predict crisis?
```

Actual Design:

```
Single snapshot (2022)
No follow-up
No crisis outcomes
No validation ❌
```

From Syllabus: "Empirical Methods in Corporate Finance" emphasizes:

- Panel data analysis
- Causal identification (DiD, IV, RDD)
- Dynamic models

Paper uses: Descriptive statistics + bivariate tests (1980s methodology)

SLIDE 10: METHODOLOGICAL RIGOR - The Circular Reasoning Trap

🚩 FLAW 3: Measurement Validity Paradox

Paper's Argument Chain:

Step 1: Use Z-scores to measure distress ✓
↓
Step 2: Find Z-scores are high → No distress ✓
↓
Step 3: Argue Z-scores are invalid because:

- X1 (working capital) overstates liquidity (93.5% is receivables collected in 1,485 days)
- X4 (equity/liability) understates leverage (equity is mostly subsidies, not retained earnings) ✓

↓
Step 4: Conclude hospitals ARE distressed...
...but Z-scores don't show it ?

The Problem: Can't use an invalid measure to test hypotheses (H1-H5), then declare it invalid!

Methodological Grade: WEAK-ADEQUATE (2.5/5)

SLIDE 11: EMPIRICAL RESULTS - The Good

✅ Valuable Empirical Contributions

1. Comprehensive Documentation

- Complete census: 90 hospitals (entire population)
- Detailed financial statements (2022)
- Transparent accounting adjustments (local standards → IPSAS)

2. Novel Greek Context Findings

Metric	Finding	Significance
Subsidy dependence	€1.9B subsidies / €336M net income	Only 3/90 hospitals self-financing
Receivables lag	1,485 days average collection	4+ years to convert A/R to cash
Capital structure	Equity/Liability = 10.9:1	Virtually zero long-term debt
Gender effect	Female CEOs → higher Z-scores (p=0.042)	21 female vs. 69 male CEOs
Size effect	Larger hospitals → lower Z-scores (p=-0.266)	Contradicts prior literature

3. Honest Acknowledgment

Paper explicitly states its primary tool (Z-scores) is flawed - rare scholarly honesty

SLIDE 12: EMPIRICAL RESULTS - The Problematic

⚠️ Interpretation Issues

Issue 1: Inconsistent Conclusions

- H1 (distress): "Accepted" (no distress) BUT "warrants deeper analysis" (actually distressed)
- H3 (size): Small hospitals "better" (higher Z) BUT Z-scores are "inflated/misleading"

Issue 2: Causal Claims Without Causal Design

CEO Gender Finding:

- Female-led hospitals: Higher Z-scores (p=0.042)
- Interpreted as: Female leadership → Better financial management
- Alternative explanations (not ruled out):
 - Women appointed to already-healthier hospitals (selection)
 - Women sort into easier-to-manage hospitals (self-selection)
 - Omitted variable (e.g., board quality) correlated with CEO gender

No controls for: CEO tenure, age, education, hospital prior performance, appointment mechanism

Issue 3: Unexplained Anomalies

- Small hospitals outperform large (contradicts economies of scale)

- Rural outperform urban (contradicts prior literature)
- Specialty has NO effect (unexpected)

Results Grade: ADEQUATE (3/5) - Good description, weak interpretation

SLIDE 13: CONTRIBUTION TO KNOWLEDGE

Contribution Matrix

Type	Claimed	Actual	Gap
Theoretical	"Redefine financial distress for public entities"	Identifies need; provides no redefinition	★ ★ ☆ ☆ ☆
Empirical	Comprehensive Greek hospital data	Good documentation; limited generalizability	★ ★ ★ ☆ ☆
Methodological	Apply Z-scores to public hospitals	Confirms known model failure; no new method	★ ★ ☆ ☆ ☆
Practical	Inform hospital oversight policy	Some actionable insights; lacks specificity	★ ★ ★ ☆ ☆

What Would Constitute a Strong Contribution?

Theoretical:

- Formal model of "transferred distress" with testable predictions
- Extension of soft budget constraint theory to healthcare
- Application of real options to government hospital decisions

Empirical:

- Panel data showing which hospitals experience crisis
- Comparative design (Greece vs. other countries; public vs. private)
- Validation of alternative distress measure

Methodological:

- New financial distress model for public entities
- Causal identification strategy (DiD, IV)
- Integration of clinical and financial outcomes

Contribution Grade: WEAK-ADEQUATE (2.5/5)

SLIDE 14: SYLLABUS ALIGNMENT - Topic Coverage

Alignment with Advanced Corporate Finance Content

Syllabus Topic	Coverage	Grade	Evidence
1. Capital Structure Theories	★ ★ ☆ ☆ ☆	Weak	Describes CS but doesn't explain it using theory; misses pecking order/trade-off applications
2. Corporate Governance	★ ★ ★ ☆ ☆	Adequate	Tests governance-performance link (gender, size) but no board/compensation data
3. Market for Corporate Control	☆ ☆ ☆ ☆ ☆	None	Mentions hospital mergers exist but no analysis; huge missed opportunity
4. International Corporate Finance	★ ★ ☆ ☆ ☆	Weak	Single-country study; no comparative institutional analysis
5. Corporate Diversification & Real Options	☆ ☆ ☆ ☆ ☆	None	Could apply real options to gov't decisions; diversification to specialty; doesn't

Methodological Standards

Syllabus Expects: "Empirical methods in corporate finance" (GMM, panel data, identification)

Paper Provides: Descriptive stats + non-parametric tests (dated approach)

Overall Alignment: ADEQUATE (3/5) - Covers 40% of syllabus topics; methodologically dated

SLIDE 15: COMPREHENSIVE STRENGTHS vs. WEAKNESSES

STRENGTHS (What the Paper Does Well)

Strength	Impact
1. Identifies important puzzle	Traditional finance tools fail for public entities - real problem
2. Comprehensive empirical documentation	Complete data on 90 hospitals; valuable for Greek policymakers

Strength	Impact
3. Honest about limitations	Explicitly acknowledges Z-scores are flawed (rare scholarly integrity)
4. Novel findings	1,485-day receivables lag, subsidy dependence quantified
5. Gender diversity evidence	Contributes to corporate governance literature on female leadership
6. Policy relevance	Identifies actionable problems (payment delays, subsidy transparency)
7. Clear writing	Well-structured, appropriate citations, transparent methods

✗ WEAKNESSES (Critical Deficiencies)

Weakness	Consequence
1. Circular reasoning	Uses invalid measure to test hypotheses, then declares it invalid
2. No theoretical development	Calls for "redefinition" but provides none - unfulfilled promise
3. Wrong research design	Cross-sectional for dynamic phenomenon; no validation possible
4. Limited causal inference	Bivariate tests only; no controls; can't distinguish causation from correlation
5. Atheoretical governance variables	CEO gender, size as crude proxies; no governance mechanisms measured
6. No engagement with CS theory	Describes extreme capital structure but doesn't explain it theoretically
7. Narrow generalizability	Single country, single year; unclear if findings transfer
8. Methodologically dated	1980s approach (descriptive) vs. modern empirical finance (causal identification)

SLIDE 16: HOW TO TRANSFORM THIS FROM C+ TO A-

🔧 MAJOR REVISIONS NEEDED

1. THEORETICAL DEVELOPMENT (Most Critical)

Current: "Financial distress needs redefinition" (stated repeatedly, never delivered)

Needed:

Step 1: Define "mission-critical distress" for public hospitals

- Inability to provide safe, accessible care
- Workforce exodus
- Quality deterioration

Step 2: Develop theoretical model

- Soft budget constraint framework
- Multi-stakeholder distress distribution
- Government intervention triggers

Step 3: Derive testable predictions

- When does transferred distress become unsustainable?
- What metrics predict government intervention?

Theoretical Frameworks to Apply:

- Kornai's soft budget constraint theory
- Agency theory (principal = state/citizens; agent = hospital managers)
- Pecking order theory (why hospitals use subsidies not debt)
- Real options (government's options to restructure/close/privatize)

SLIDE 17: TRANSFORMATION ROADMAP (Continued)

2. RESEARCH DESIGN OVERHAUL

Current: Single-year cross-section with bivariate tests

Needed:

Option A: Panel Data Design (Preferred)

- Collect 5-10 years of data
- Implement fixed effects regression
- Model: $Z\text{-score}(it) = \alpha + \beta \cdot \text{Governance}(it) + \text{Hospital}(i) + \text{Year}(t) + \varepsilon$
- Controls for time-invariant hospital characteristics

Option B: Quasi-Experimental Design

- Identify policy shock (e.g., 2024 ULS integration reform)
- Difference-in-differences:
 - Treatment: Hospitals merged into ULS
 - Control: Hospitals not yet merged
 - Measure: Financial metrics before/after

Option C: Comparative Design

- Compare Greek hospitals to:
 - Portuguese SNS (similar system, different outcomes)
 - Greek private hospitals (market discipline effect)
 - Pre/post economic crisis (2008-2015)

SLIDE 18: TRANSFORMATION ROADMAP (Continued)

3. DEVELOP ALTERNATIVE DISTRESS MEASURE

Current: Relies solely on Z-scores, admits they're invalid

Needed: "Public Hospital Financial Sustainability Index (PHFSI)"

Components (equally weighted):

1. Operational Self-Sufficiency Ratio
= Operating Revenue / (Operating Expenses - Subsidies)
2. Stakeholder Pressure Index
= (Accounts Payable Days / 90) + (Overdue Liabilities / Total Liabilities)
3. Receivables Realization Rate
= Cash Collections / Accrued Revenue (adjusted for lag)
4. Subsidy Dependency Ratio
= State Subsidies / Total Revenue
5. Clinical Quality Maintenance Index
= (Patient Safety Score + Access Metrics) / Benchmarks

Validate by:

- Showing low PHFSI predicts government intervention
- Demonstrating PHFSI captures problems Z-scores miss
- Testing across multiple countries/time periods

SLIDE 19: TRANSFORMATION ROADMAP (Continued)

4. STRENGTHEN GOVERNANCE ANALYSIS

Current: Binary CEO gender, hospital size (atheoretical)

Needed: Comprehensive Governance Framework

Governance Dimension	Measurement	Theoretical Grounding
Board Structure	Size, independence, medical vs. administrative expertise	Agency theory: monitoring effectiveness
Executive Compensation	Fixed vs. performance-based, KPIs	Incentive alignment theory
Political Connections	Manager appointment process, political affiliation	Public choice theory
Stakeholder Representation	Patient/community board seats	Stakeholder theory
Transparency Mechanisms	Public reporting quality, audit frequency	Information asymmetry theory

Analysis:

- Multivariate regression with full controls
- Mediation analysis: Does governance affect distress through financial decisions?
- Interaction effects: Does governance matter more in certain contexts?

SLIDE 20: REVISED CONTRIBUTION STATEMENT

Original vs. Enhanced Contribution

ORIGINAL PAPER CLAIMS:

"This study paves the way for redefining financial distress in public healthcare organizations"

Actually delivers: Documentation that Z-scores don't work (already known from soft budget constraint literature)

ENHANCED PAPER WOULD CLAIM:

Theoretical Contribution:

"We develop a stakeholder-distributed financial distress framework for mission-driven organizations operating under soft budget constraints, demonstrating that traditional bankruptcy-based models mischaracterize distress as transferred to suppliers, taxpayers, and service recipients rather than borne by the entity itself."

Empirical Contribution:

"Using longitudinal data from 90 Greek public hospitals (2017-2024), we validate a new financial sustainability measure that predicts government intervention with 85% accuracy, outperforming traditional Z-scores which show no predictive validity (AUC=0.51 vs. 0.89)."

Methodological Contribution:

"We introduce the Public Hospital Financial Sustainability Index (PHFSI), a multi-dimensional measure integrating operational self-sufficiency, stakeholder pressure, and clinical quality maintenance, providing a replicable tool for comparative health systems research."

SLIDE 21: GRADING JUSTIFICATION

Final Grade Calculation

Dimension	Weight	Score	Weighted
Theoretical Foundation	25%	40/100	10.0
Methodological Rigor	30%	45/100	13.5
Results & Interpretation	20%	60/100	12.0
Contribution to Knowledge	15%	50/100	7.5
Course Alignment	5%	60/100	3.0
Practical Implications	3%	70/100	2.1
Limitations & Future Research	2%	60/100	1.2
TOTAL	100%		49.3/100

With Writing Quality Bonus: +20 points (well-written, clear, honest)

FINAL GRADE: 69.3/100 = C+ / B-

What This Grade Means

C+ / B- = "Adequate with Significant Limitations"

 **Pass Standards:** Competent empirical work, identifies real problem, honest scholarship

 **Falls Short Of:** Theoretical development, methodological rigor, causal inference, comprehensive contribution expected in Advanced Corporate Finance

Appropriate for: Health management journals (current: *Frontiers in Public Health*, Q2)

Not ready for: Q1 finance/management journals (*Journal of Finance*, *Strategic Management Journal*, *Journal of Health Economics*)

SLIDE 22: APPROPRIATE USE IN ADVANCED CF COURSE

Pedagogical Value: HIGH Despite Grade

This Paper is Excellent Teaching Material For:

1. Discussion Topics:

- When do corporate finance tools fail?
- How does institutional context matter?
- What is "distress" when bankruptcy isn't possible?
- How do soft budget constraints change everything?

2. Methodological Lessons:

- Dangers of applying models outside their design context
- Importance of validation samples
- Cross-sectional vs. longitudinal design choices
- Circular reasoning traps

3. Theoretical Extensions:

- How to adapt capital structure theory to public entities
- Agency problems in government-owned organizations
- Real options in government decision-making
- Stakeholder theory in mission-driven organizations

💡 Suggested Class Structure (90 min)

Time	Activity
0-15 min	Present paper summary and main findings
15-30 min	Critical analysis: What's wrong with using Z-scores here?
30-50 min	Small group work: Design a better study (theory, design, measures)
50-70 min	Group presentations: Each group presents their improved approach
70-90 min	Synthesis: What did we learn about applying CF to non-standard contexts?

Learning Outcomes:

-  Critical evaluation of empirical research
-  Understanding model assumptions and limitations
-  Translating theory to non-corporate contexts
-  Research design thinking

SLIDE 23: COMPARISON TO PROJECT EXEMPLARS

 How This Paper Compares to Gemini/ChatGPT/Claude Proposals

Dimension	Karakolias Paper	Gemini/ChatGPT/Claude
Theoretical Depth	☆☆☆☆ Identifies problem, no solution	★★★★★ Formal frameworks (real options, SROI, data valuation)
Research Design	☆☆☆☆ Cross-sectional, descriptive	★★★★★ Quasi-experimental, causal inference
CF Theory Integration	☆☆☆☆ Limited engagement	★★★★★ Deep integration (capital budgeting, cost of capital, structured finance)
Innovation	☆☆☆☆ Applies existing model	★★★★★ Creates new instruments (CVBs, VB receivables financing)
Data Sophistication	☆☆☆☆ Financial statements	★★★★★ Multi-source integration (clinical + financial + outcomes)
Actionability	☆☆☆☆ General recommendations	★★★★★ Specific mechanisms (SPVs, tranching, governance committees)

 Key Lessons from Comparison

What Makes Q1-Level Research:

1. **Theoretical Ambition:** Don't just identify problems → Solve them with formal models
2. **Methodological Rigor:** Design to establish causality, not just document patterns
3. **Integration:** Bridge multiple literatures (finance + healthcare + institutional economics)
4. **Innovation:** Create new tools/methods/measures, don't just apply existing ones
5. **Specificity:** Concrete mechanisms, not vague recommendations

Karakolias stopped at step 1; exemplar proposals complete all 5.

SLIDE 24: KEY TAKEAWAYS - The Five Critical Lessons

 1. CONTEXT MATTERS - Model Validity is Conditional

Lesson: Altman's Z-score works for private firms in competitive markets with bankruptcy risk. It **fails catastrophically** when:

- Bankruptcy is impossible (government guarantee)
- "Equity" is subsidies, not retained earnings

- "Liabilities" exclude debt (no access to capital markets)
- "Working capital" is uncollectible receivables

Implication: Always interrogate model assumptions before application

2. DESCRIPTION ≠ EXPLANATION

Paper does: Documents patterns (size matters, gender matters, location matters)

Paper doesn't do: Explains WHY using theory

Lesson: Advanced CF requires **causal mechanisms**, not just correlations

3. CIRCULAR REASONING TRAP

The Logic Flaw:

```
"We'll measure X with tool Y"
↓
"Tool Y says no X exists"
↓
"But tool Y is invalid"
↓
"Therefore X exists" ⚠️ INVALID!
```

Lesson: Can't use an invalid measure to support claims—must develop valid alternative FIRST

4. VALIDATION REQUIRES OUTCOMES

Prediction models need:

- Sample with both successes and failures
- Observation period to see predictions realized
- Clear definition of outcome being predicted

This paper has: Zero failures, one time point, unclear outcome definition

Lesson: Can't validate without variation in the thing you're predicting

5. THEORETICAL PROMISE vs. DELIVERY

Claiming to "redefine" something requires:

1. Clear new definition
2. Theoretical justification
3. Operational measures
4. Empirical validation

This paper provides: Only #1 (partially)

Lesson: Don't overpromise theoretical contributions; deliver what you claim or claim what you deliver

SLIDE 25: DISCUSSION QUESTIONS FOR CLASS

Opening Questions (Warm-up)

1. **First Impressions:** After reading the abstract and introduction, what did you expect this paper to contribute? Did it deliver?
 2. **The Z-Score Question:** Is it ever appropriate to apply a bankruptcy prediction model to organizations that cannot go bankrupt? Why or why not?
-

Theoretical Questions (Core Issues)

3. **Defining Distress:** If bankruptcy isn't the outcome, what SHOULD "financial distress" mean for a public hospital? How would you define it?
4. **Transferred Distress:** The paper argues distress is "transferred" to suppliers and taxpayers. Is this a useful concept? Can we model it formally?
5. **Capital Structure Puzzle:** Why do Greek public hospitals have equity/liability ratios of 10.9:1 (almost no debt)? Which capital structure theories explain this: pecking order, trade-off, market timing, or something else?

6. **Agency Problems:** Who is the "principal" and who is the "agent" in a public hospital? What are the specific agency conflicts, and how do they differ from private corporations?

Methodological Questions (Research Design)

7. **Research Design:** If you were redesigning this study, what would you change first: the measure (Z-score), the design (cross-sectional), the sample (Greece only), or something else?
8. **Causality Challenge:** The paper finds female CEOs lead hospitals with higher Z-scores. What are three alternative explanations (other than "female leadership causes better financial management") that could explain this correlation?
9. **Validation Dilemma:** How can you validate a financial distress model when the population has zero failures? Is this a fatal flaw or a solvable problem?

Application Questions (Real-World)

10. **Policy Implications:** If you were the Greek Minister of Health, which finding from this paper would you act on first, and how?
11. **International Transfer:** Would the findings from Greek hospitals apply to:
- NHS England? Why or why not?
 - Portuguese SNS? Why or why not?
 - US public hospitals? Why or why not?
12. **Private Sector Analogy:** Can you think of situations in private corporations where traditional financial distress models would similarly fail? (e.g., too-big-to-fail banks, companies with implicit government guarantees)

Advanced Questions (Extensions)

13. **Real Options Application:** How could you apply real options theory to model the government's decision about when to intervene in a struggling public hospital? What are the options, the uncertainties, and the exercise triggers?
14. **Structured Finance Solution:** Design a financial instrument that could help public hospitals manage the cash flow gap between providing care and receiving payment from insurance funds (1,485-day lag). Consider the project proposals' ideas about value-based receivables financing.
15. **Alternative Performance Metrics:** If Z-scores don't work, what SHOULD we measure to assess public hospital financial health? Design a new index with 3-5 components and justify each theoretically.

SLIDE 26: SELF-ASSESSMENT RUBRIC FOR STUDENTS

Use This to Evaluate YOUR Own Research

Rate Your Paper (1-5 on each dimension):

Question	1 = Poor	5 = Excellent	This Paper's Score
Theory: Do I develop/extend theory or just apply existing tools?	Apply only	Original theory	2
Design: Can my design support causal claims or only correlations?	Correlation only	Causal ID	2
Measures: Are my constructs valid for my specific context?	Context-inappropriate	Validated for context	2
Contribution: Do I promise what I deliver and deliver what I promise?	Gap between claim & delivery	Perfect alignment	2
Methods: Do I use state-of-the-art empirical techniques?	Dated approach	Cutting edge	2
Integration: Do I bridge multiple literatures meaningfully?	Single literature	Deep integration	2
Rigor: Would reviewers find major flaws in my logic or design?	Multiple flaws	Bulletproof	2
Writing: Is my paper clear, well-structured, and honest?	Unclear/oversold	Crystal clear	4

Scoring Guide:

- **32-40 points:** A-level work, Q1 publication potential
- **24-31 points:** B-level work, solid contribution
- **16-23 points:** C-level work, adequate but limited ← **This paper: 18/40**
- **8-15 points:** D-level work, major revisions needed
- **<8 points:** Fundamental reconceptualization required

SLIDE 27: RESOURCES FOR FURTHER LEARNING

To Understand What This Paper is Trying to Do

Soft Budget Constraints:

- Kornai, J. (1986). "The Soft Budget Constraint." *Kyklos*, 39(1): 3-30.

- Maskin, E. & Xu, C. (2001). "Soft Budget Constraint Theories." *Economics of Transition*, 9(1): 1-27.

Financial Distress in Non-Profit/Public Entities:

- Ziolo, M. et al. (2017). "The Financial Distress of Public Sector Entities." In *Risk Management in Public Administration*.
- Tuckman, H.P. & Chang, C.F. (1991). "A Methodology for Measuring Financial Vulnerability of Charitable Nonprofits." *Nonprofit and Voluntary Sector Quarterly*, 20(4): 445-460.



To Learn How to Do This BETTER

Empirical Corporate Finance Methods:

- Roberts, M.R. & Whited, T.M. (2013). "Endogeneity in Empirical Corporate Finance." In *Handbook of the Economics of Finance*, Vol. 2.
- Angrist, J.D. & Pischke, J.S. (2009). *Mostly Harmless Econometrics: An Empiricist's Companion*.

Capital Structure in Unusual Contexts:

- Becker, B. & Strömberg, P. (2012). "Fiduciary Duties and Equity-Debtholder Conflicts." *Review of Financial Studies*, 25(6): 1931-1969.

Public Sector Financial Management:

- Goddard, A. & Powell, J. (1994). "Accountability and Accounting: Using Naturalistic Methodology to Enhance Organizational Control." *Accounting, Auditing & Accountability Journal*, 7(2): 50-69.

Real Options in Healthcare:

- Balasubramanian, N. et al. (2010). "Real Options in Healthcare." In *Handbook of Quantitative Finance and Risk Management*.

SLIDE 28: FINAL VERDICT - When is This Paper Useful?



USE THIS PAPER WHEN:

Teaching Context:

- Illustrating limits of financial models across institutional contexts
- Discussing soft budget constraints and government ownership
- Case study in research design challenges
- Example of honest scholarly acknowledgment of limitations

Practice Context:

- Understanding Greek healthcare system specifically
- Documenting payment delay problems in public systems
- Policy discussions about transparency in subsidy-dependent entities

Research Context:

- Motivation for developing better public sector financial metrics
- Baseline data for longitudinal studies of Greek hospitals
- Comparison point for other countries' public hospital finances



DON'T USE THIS PAPER FOR:

- **Theoretical guidance** on financial distress in public entities (theory not developed)
- **Methodological template** for empirical CF research (design inadequate)
- **Causal claims** about governance effects (no causal identification)
- **Generalizable conclusions** about public hospitals (single country, single year)
- **Validation** of Z-scores in public context (paper shows they don't work)
- **Q1 journal standards** for student thesis/dissertation (would need major revision)



BOTTOM LINE

This paper is:

- A good *starting point* for a research program
- A valuable *data documentation* exercise
- An honest *problem identification* study
- A *missed opportunity* for theoretical development

It is NOT:

- A complete solution to the problem it identifies
- A methodologically rigorous causal study
- A significant theoretical contribution to corporate finance
- Ready for publication in Q1 finance/management journals

Grade: C+/B- (70-75/100) - Adequate empirical work with important limitations

SLIDE 29: YOUR ASSIGNMENT - Critical Review Practice

Individual Exercise (Optional Follow-up)

Task: Write a 2-page referee report on this paper as if you were reviewing it for:

Option A: *Journal of Health Economics* (Q1, health economics focus)

Option B: *Strategic Management Journal* (Q1, strategy/governance focus)

Option C: *Frontiers in Public Health* (Q2, where it's actually published)

Your Report Should Include:

1. **Summary (1 paragraph):** What does the paper do?
2. **Strengths (3-4 bullets):** What's good about it?
3. **Major Concerns (3-4 numbered points):** What needs major revision?
4. **Minor Issues (3-4 bullets):** What needs minor improvement?
5. **Recommendation:**
 - Accept as is
 - Minor revisions
 - Major revisions (revise & resubmit)
 - Reject but encourage resubmission
 - Reject

Guiding Question: How would your recommendation differ across the three journals, and why?

SLIDE 30: CONCLUSION - What We Learned

Meta-Lessons from This Assessment

About Research:

1. **Context is king** - Models valid in one setting may fail in another
2. **Theory matters** - Description without explanation is insufficient
3. **Design determines what you can claim** - Cross-sectional → correlation only
4. **Validation requires outcomes** - Need variation to test predictions
5. **Honesty is valuable** - Acknowledging limitations is scholarly integrity

About Academic Writing: 6. **Promise what you deliver** - Don't claim theoretical contributions you don't provide 7. **Circular reasoning is fatal** - Can't use invalid measure to validate claims 8. **Integration adds value** - Bridge literatures, don't just apply tools

About Advanced Corporate Finance: 9. **Rigor has multiple dimensions** - Theoretical, empirical, methodological 10. **Standards evolve** - Methods adequate in 1990s may be insufficient today

The Bigger Picture

This paper represents **honest, competent work that falls short of excellence** - a common situation in academic research.

The valuable lesson: Understanding *why* it falls short (theory, design, causality) teaches us what *excellence requires*.

Your goal as researchers:

- Learn from both strengths (comprehensive data, honest assessment)
 - And weaknesses (limited theory, inadequate design)
 - To produce work that advances knowledge meaningfully
-

THANK YOU

Questions? Discussion?

APPENDIX SLIDES (Optional - For Detailed Discussion)

A1: Detailed Z-Score Formula Breakdown

Altman's Z"-Score Model:

Where:

- (Liquidity)
- (Profitability History)
- (Operating Efficiency)
- (Leverage/Solvency)

Cutoff Interpretation:

- **Z < 1.10:** Distress zone (high bankruptcy probability)
- **1.10 ≤ Z ≤ 2.60:** Gray zone (uncertain)
- **Z > 2.60:** Safe zone (low bankruptcy probability)

Greek Hospital Results:

- Mean Z = 15.7 (6× safe threshold)
- Min Z = 5.5 (still 2× safe threshold)
- All 90 hospitals in "safe zone"

But Why?

- X1 contribution: $6.56 \times 0.617 = \mathbf{4.05 \text{ points}}$
- X4 contribution: $1.05 \times 10.9 = \mathbf{11.45 \text{ points}}$
- Together: **15.5 / 15.7 = 99% of score**

A2: Comparison of Empirical Methods - Timeline

EVOLUTION OF EMPIRICAL CORPORATE FINANCE METHODS

1960s-1970s: UNIVARIATE & DESCRIPTIVE

- Financial ratio analysis
- Mean comparisons, t-tests

→ Altman (1968) original Z-score

1980s-1990s: MULTIVARIATE & CROSS-SECTIONAL

- OLS regression
- Logit/probit models
- Non-parametric tests (Mann-Whitney, Kruskal-Wallis)

→ This paper's methodology ★

2000s: PANEL DATA & FIXED EFFECTS

- Control for unobserved heterogeneity
- Difference-in-differences
- Arellano-Bond dynamic panel

2010s-Present: CAUSAL IDENTIFICATION

- Regression discontinuity designs
- Instrumental variables
- Natural experiments, staggered DiD
- Machine learning for prediction
- Structural models

→ Expected in Q1 journals today ★

Verdict: Paper uses 1980s-1990s methodology in 2025 - dated by 30+ years

A3: Capital Structure Theory - Application to Public Hospitals

Pecking Order Theory (Myers 1984):

Prediction: Firms prefer internal financing > debt > equity (due to information asymmetry)

Application to Public Hospitals:

Preference Hierarchy:

1. Government subsidies (analogous to "internal" from state owner)

2. Retained earnings (very limited)

- 3. Debt (NOT USED – would require market discipline)
- 4. Equity issuance (IMPOSSIBLE – no shares)

Result: Zero leverage (observed)

Trade-Off Theory (Kraus & Litzenberger 1973):

Prediction: Optimal leverage balances tax shields vs. financial distress costs

Application to Public Hospitals:

Tax Shield Benefits: ZERO (public entities don't pay taxes)
Financial Distress Costs: VERY HIGH (threatens public mission)

Optimal Leverage: ZERO ✓

Result: No debt (observed)

Conclusion: Both theories predict the observed capital structure!

Paper's Missed Opportunity: Could have used theory to explain, not just describe

A4: Statistical Power Analysis - Example

H4: Hospital Specialty Affects Z-scores

Sample Composition:

- General hospitals: n=68
- General + health centers: n=7
- University hospitals: n=7
- Anticancer hospitals: n=2 ⚠
- Other: n=6

Kruskal-Wallis Test Result: $H = 2.275$, $p = 0.685$ (not significant)

But: With only 2 anticancer hospitals, statistical power to detect a true difference is:

Interpretation: 85% chance of Type II error (false negative)

Conclusion: Null finding may reflect inadequate power, not absence of effect

Paper's Flaw: Accepts null hypothesis without discussing power

A5: Recommended Reading - Corporate Finance Classics

For Students Wanting to Understand Advanced CF Standards:

1. **Berk & DeMarzo** (2014). *Corporate Finance*, 3rd ed. - Textbook foundation
2. **Myers, S.C.** (1984). "The Capital Structure Puzzle." *Journal of Finance*, 39(3): 575-592. - Pecking order theory
3. **Jensen, M.C.** (1986). "Agency Costs of Free Cash Flow." *American Economic Review*, 76(2): 323-329. - Agency theory
4. **Fama & French** (2002). "Testing Trade-Off and Pecking Order Predictions About Dividends and Debt." *Review of Financial Studies*, 15(1): 1-33. - Empirical testing
5. **Roberts & Whited** (2013). "Endogeneity in Empirical Corporate Finance." *Handbook of the Economics of Finance*, Vol. 2, Ch. 7. - Methods

For Understanding Public Sector Finance:

6. **Kornai, J.** (1980). *Economics of Shortage*. - Soft budget constraints
7. **Shleifer, A.** (1998). "State versus Private Ownership." *Journal of Economic Perspectives*, 12(4): 133-150. - Government ownership

END OF PRESENTATION

Total Slides: 30 main + 5 appendix = 35 slides

Estimated Presentation Time:

- Full version: 60-75 minutes
- Core version (slides 1-24): 45-50 minutes
- Summary version (slides 1-5, 15-17, 24, 28-30): 25-30 minutes

Recommended Format:

- Present slides 1-24 in class
- Use slides 25-27 for discussion

- Provide slides 28-30 and appendix as handout
-

Suggested Visual Enhancements

For Maximum Impact, Add:

1. Color Coding:

- Red boxes for weaknesses/problems
- Green boxes for strengths
- Yellow boxes for mixed/adequate
- Blue boxes for theoretical concepts

2. Icons/Symbols:

-  for critical flaws
-  for strengths
-  for insights
-  for data/empirics
-  for key takeaways

3. Graphs/Charts:

- Spider diagram showing scores across 8 dimensions
- Timeline of empirical methods evolution
- Comparison bar chart (this paper vs. exemplars)

4. Flow Diagrams:

- Research design comparison (current vs. ideal)
- Circular reasoning visualization
- Transferred distress stakeholder map

This presentation structure provides a comprehensive, visually organized critique suitable for classroom presentation in an Advanced Corporate Finance course. Would you like me to elaborate on any specific section or create additional supporting materials?